

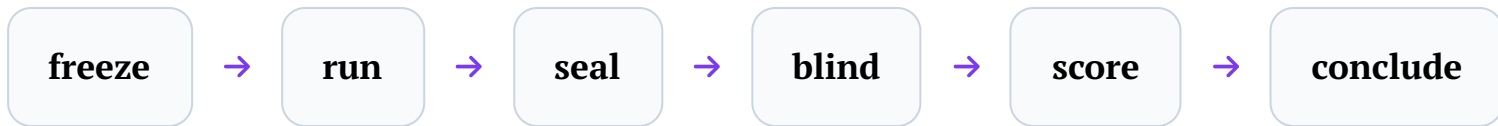
The Experiment Fought Back

Trials, failures and useful negative results

The engineering story between “let’s test it” and “we can trust the test.”



The plan looked linear



Reality

fail → inspect → preserve → repair → rerun → doubt → audit → narrow the claim

Five runs and a harness rewrite

04:23

Schema failure

0 observations

04:27

Haiku smoke

81 / 81

08:15

Gemma partial

discarded

08:31

Gemma clean

81 / 81

09:00

Persona pairs

60 / 60

12:23

Go + Seatbelt

fail closed

Every arrow represents a changed threat model, not just another command.

Failure 1: no model was called

Expected

81 structured Claude responses

Observed

72 local schema errors

9 blocked fusers

0 inference responses

`$schema` remained in source and provenance, but was omitted from the CLI argument.

The failed run stayed sealed. It never became data.

Success created a new problem

0.81–0.86

Mean F1 for most final Haiku conditions

Plumbing

worked

Headroom

did not

Verdict

not available

A frontier-ish model on easy synthetic packets mostly measured ceiling effects and one decoding draw.

Metric key — because nobody remembers F1

Key	Meaning
TP	one unique planted defect correctly found
FP	one unique unsupported claim after semantic de-duplication
FN	one planted defect missed
Precision	$TP / (TP + FP)$ — how much of the review was supported
Recall	$TP / (TP + FN)$ — how much of the defect set was found
F1	$2TP / (2TP + FP + FN)$ — balance of precision and recall

Equal F1 means equal counts. It does not mean equal defects, remedies or severity.

Equal scores hid different reviews

S1

found RD-04

missed RD-06

same F1

M0

found RD-06

missed RD-04

Count metric

Same TP/FP/FN balance

Coverage metric

Different defect identity and contribution

The audit found four quiet biases

Order

A set discarded the seeded schedule.

False positives

Seven repeats of one bad claim counted seven times.

Fusion

A capable arbiter flattened panel differences.

Sampling

One output could not identify a prompt effect.

Repairs: preserve order · cluster false claims · publish raw unions · run paired repeats

Failure 2: the prompt looked like an answer



Partial run

71 success · 7 quota failures · 3 blocked fusers

Repair

assistant events only · provider errors
promoted · retries + backoff

Sealed. Discarded before rating. Replaced from a new seed.

The rater broke too

Schema dialect

Vertex required explicit types beside enums.

Wrong JSON object

The extractor selected the embedded schema's `judgements` property.

Accept the requested root only when its value is an array.

419 captured, schema-valid ratings were recovered mechanically.
No replacement model call. No changed judgement.

Cheap model, useful spread

Arm	Final F1	Recall
S0 generic	0.819	0.708
M0 repeated omnibus	0.863	0.792
M1 functional specialists	0.911	0.875
M2 fictional specialists	0.857	0.792

Before fusion

M1 recall 0.917 vs M0 0.792

Still easy

S0 remained above 0.81

The council exposed more signal—and more arbitration work.

The fictional overlay lost this round

Functional correctness

0.790

mean F1

K-2SO wrapper

0.748

mean F1

`fictional - functional = -0.0425`

2 fictional wins · 16 ties · 12 functional wins

Names can remain mnemonic labels. The prose still has to earn its tokens.

Isolation had to become executable

Before

- Python controller
- provider child inside clean worktree
- correct flags were the main boundary

Now

- standard-library Go harness
- mandatory macOS Seatbelt
- empty cwd + ephemeral HOME
- repository/worktree read must fail
- external CLI entrypoints frozen by digest

```
scratch write: allowed · repository read: denied
```

Authentication fought the sandbox

agy

Ephemeral HOME triggered an interactive keychain dependency.

Direct Claude CLI

Shared login state disappeared with the real HOME.

Do not weaken isolation. Reject both clients before launch; pin Anthropic, Google, OpenAI and Gemma models through sterile Pi.

Copy auth + model registry only. Never settings, skills, history, sessions or project context.

Internet is useful — and contaminating

Production goal

Use every legal tool that gets the right fix quickly.

Experimental goal

Do not find the matching upstream PR and copy its answer.

Base arm

frozen source + local search/tests

Docs arm

frozen offline corpus

Internet arm

declared factor + controlled proxy

Provider-side search remains unobservable unless the provider exposes and disables it.

The next test changed shape

PLANNED

Persona factorial

480 calls

Generalise one negative overlay result across eight roles.

?

PRIORITISED

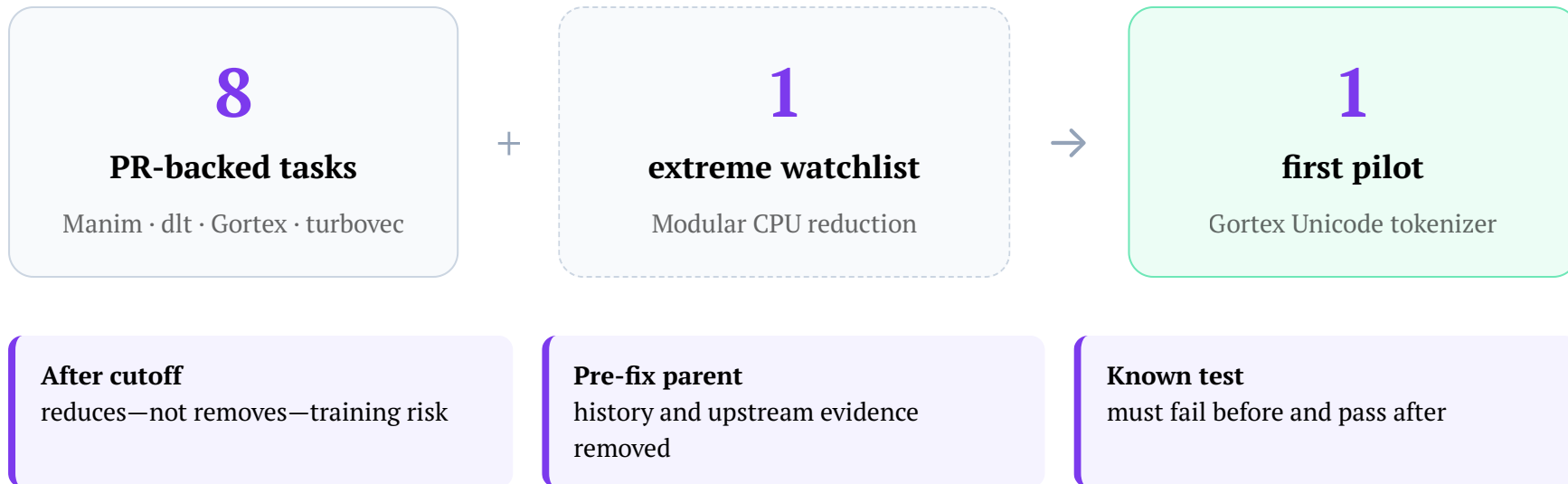
Ecological pilot

real code

First prove the mechanism survives outside planted packets.

Ceiling effects made “more synthetic calls” less urgent than “better evidence.”

Real-code curation was a funnel



Modular stayed gated: no PR review and no frozen external build closure.

The snapshot failed four different ways



Rule: preserve each attempt; repair code; start a new attempt.

Clean source is infrastructure—not a result

history-free Gortex parent

4,093

entries

37.8 MB

sanitised source

✓ tree digest 41a9a14b...

✓ .git absent

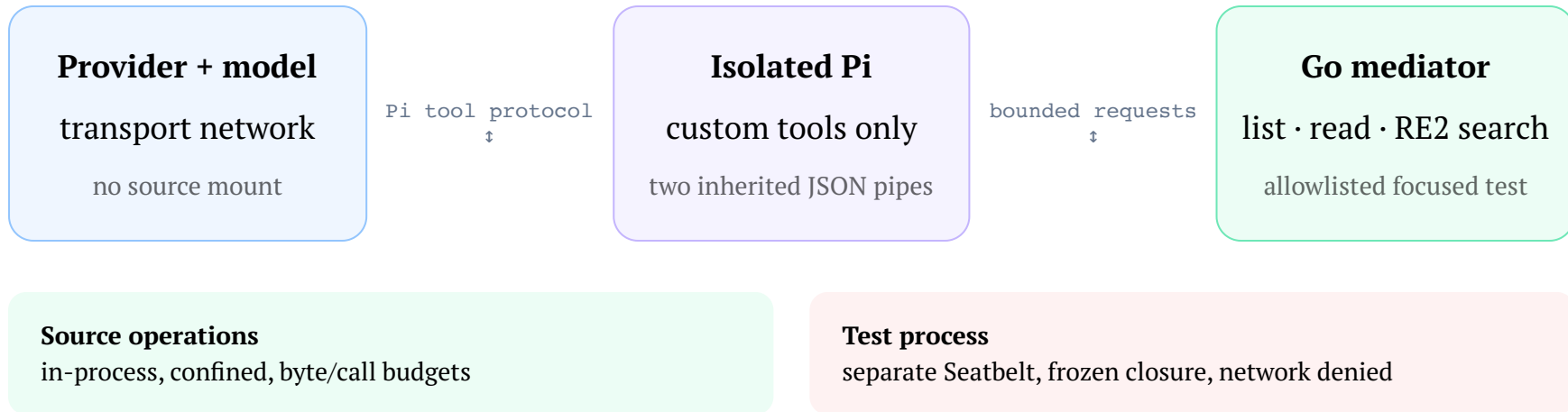
✓ parent panic reproduced

✓ evidence commit passed

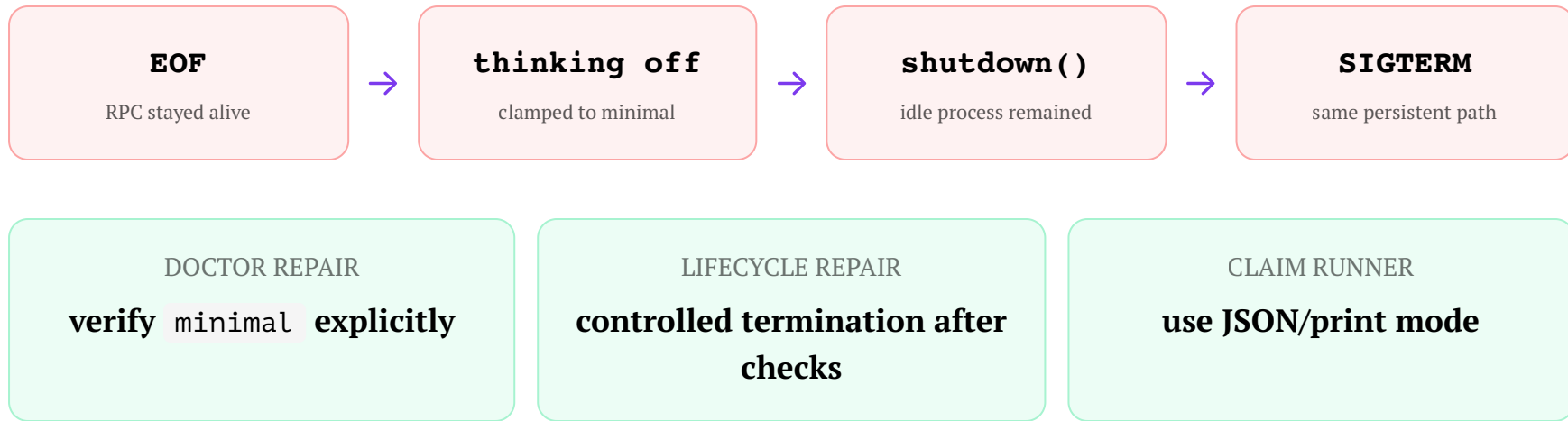
✓ source / controller / network probes

0 ecological respondent calls

Source and network could not share a process



The doctor found assumptions, not bugs



All failed attempts preserved. No prompt sent. No provider turn.

The evidence ladder now has a hard gate

1

Synthetic plumbing

complete

2

Negative overlay calibration

complete

3

History-free snapshot

sealed

4

Mediator + Pi doctor

sealed · zero calls

5

Claim runner + mock

current gate

6

Compared arms

not started

Before first call

freeze byte-level prompt factor, model, retries, endpoint and human aggregation

Then measure

supported findings, remedy quality, tool/token/cost/latency efficiency

Keep the receipts.

Failures

are methodology

Nulls

are product decisions

Negative results

remove theatre

Reproducibility

includes the pivots

The council reviews the artifact.

The experiment must review itself.

`experiment/LAB_NOTEBOOK.md` · `experiment/CONTAMINATION_REVIEW.md` · `experiment/results/`